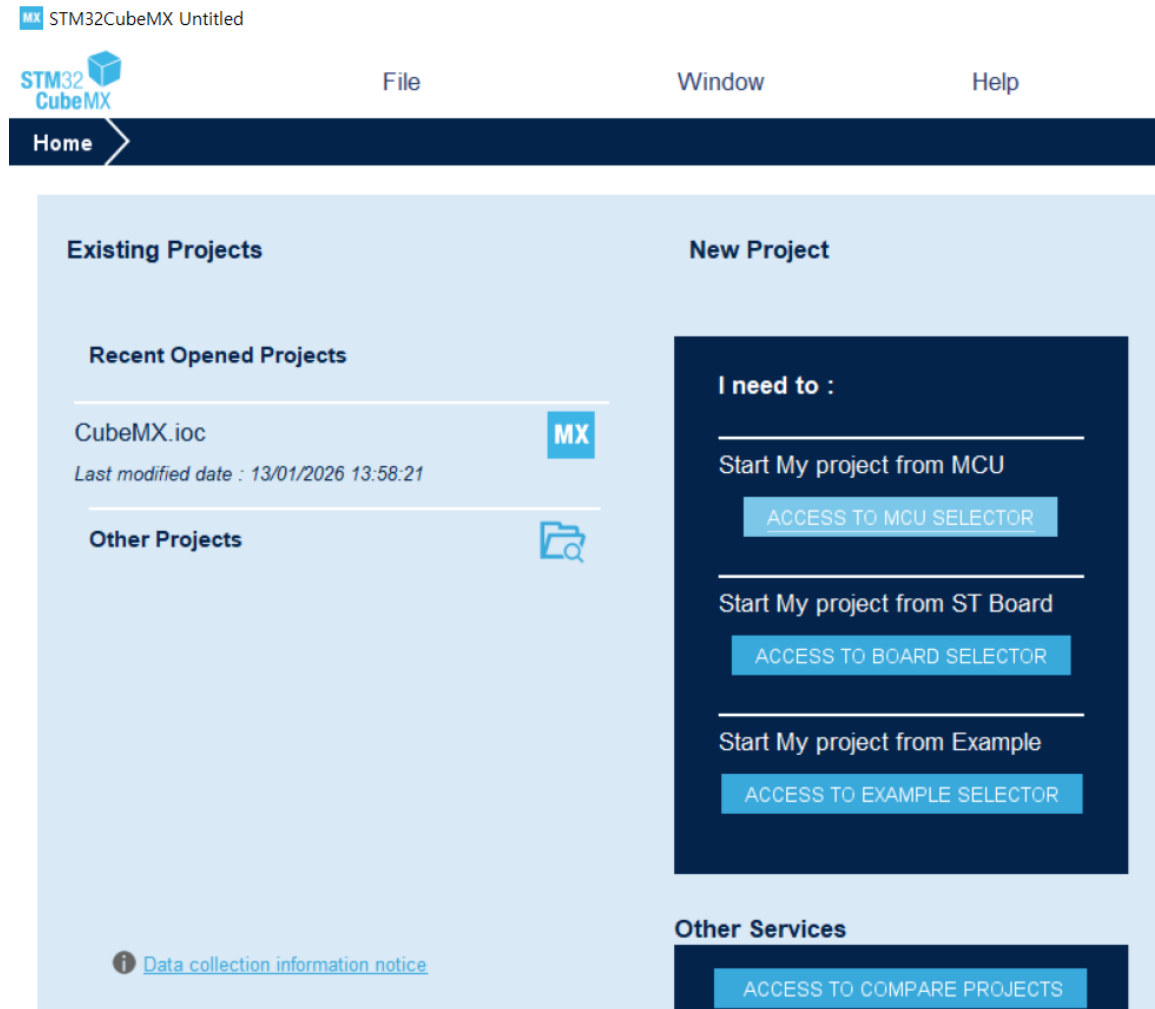


SHT20 - 습도 센서 실습

2026.06.17

Project 생성

- STM32CubeMX 실행 후 "ACCESS TO MCU SELECTOR" 선택



Project 생성

- STM32F407ZET6 선택 후 "Start Project" 클릭

MX New Project from a MCU/MPU

MCU/MPU Selector Board Selector Example Selector Cross Selector

MCU/MPU Filters

Commercial Part Number STM32F407ZET6

PRODUCT INFO

- Segment
- Series
- Line
- Marketing Status
- Price
- Package
- Core
- Coprocessor

STM32F4 Series

STM32F407ZET6

High-performance foundation line, Arm Cortex-M4 core with DSP and FPU, 512 Kbytes of Flash memory, 168 MHz CPU, ART Accelerator, Ethernet, FSMC

ACTIVE
Product is in mass production

Unit Price for 10kU (US\$): 5.605

4 mm LQFP 144 20x20x1.

The STM32F405xx and STM32F407xx family is based on the high-performance

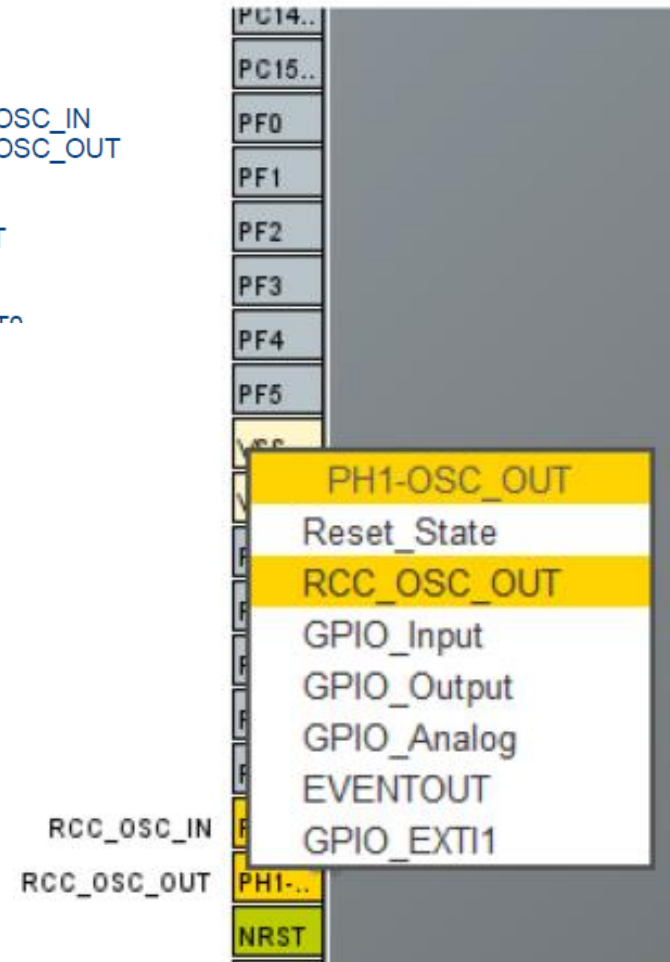
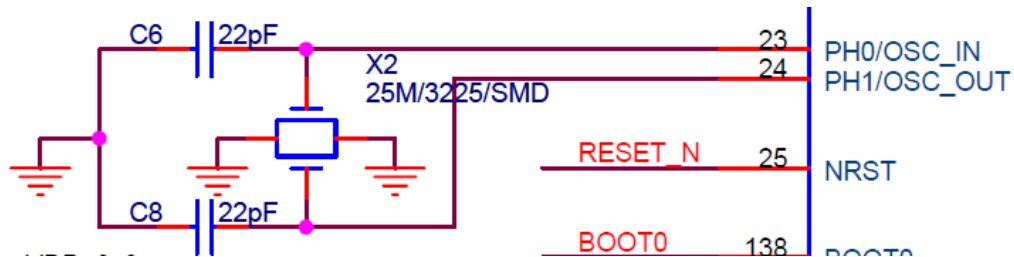
MCUs/MPUs List: 2 items

* Comm...	Part No	Reference	Mark...	Unit Pric...	B...	Package	Flash	RAM	...	Fre...
☆ STM32F4...	STM32...	STM32...	Active	5.605		LQFP 144 20x...	512 ...	192 ...	1...	168 ...
☆ STM32F...	STM32...	STM32...	Active	5.605		LQFP 144 20x...	512 ...	192 ...	1...	168 ...

Export

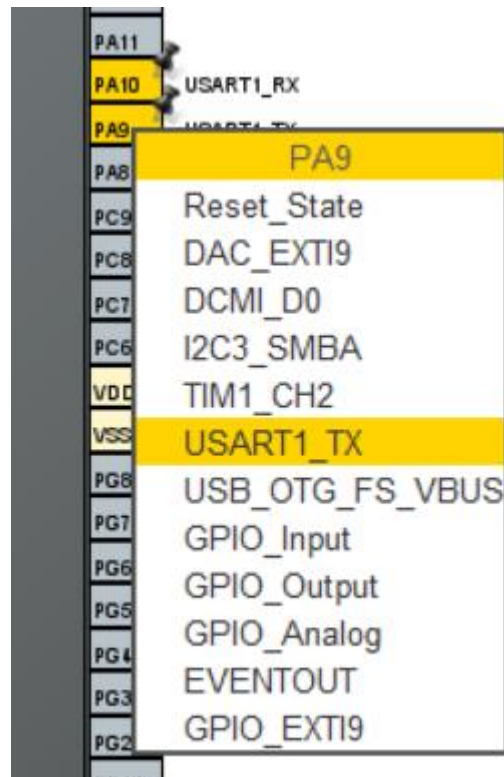
Project 생성

- HSE(25MHz) 핀 설정 - 회로도 참조
- HSE : PH0-OSC_IN, PH1-OSC_OUT



Project 생성

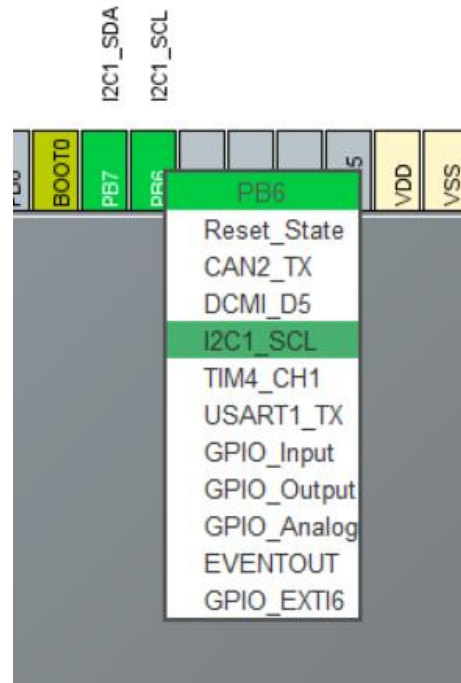
- USART1핀 설정 - 회로도 참조
- UART1 : PA9-USART1_TXD, PA10-USART1_RXD



Project 생성

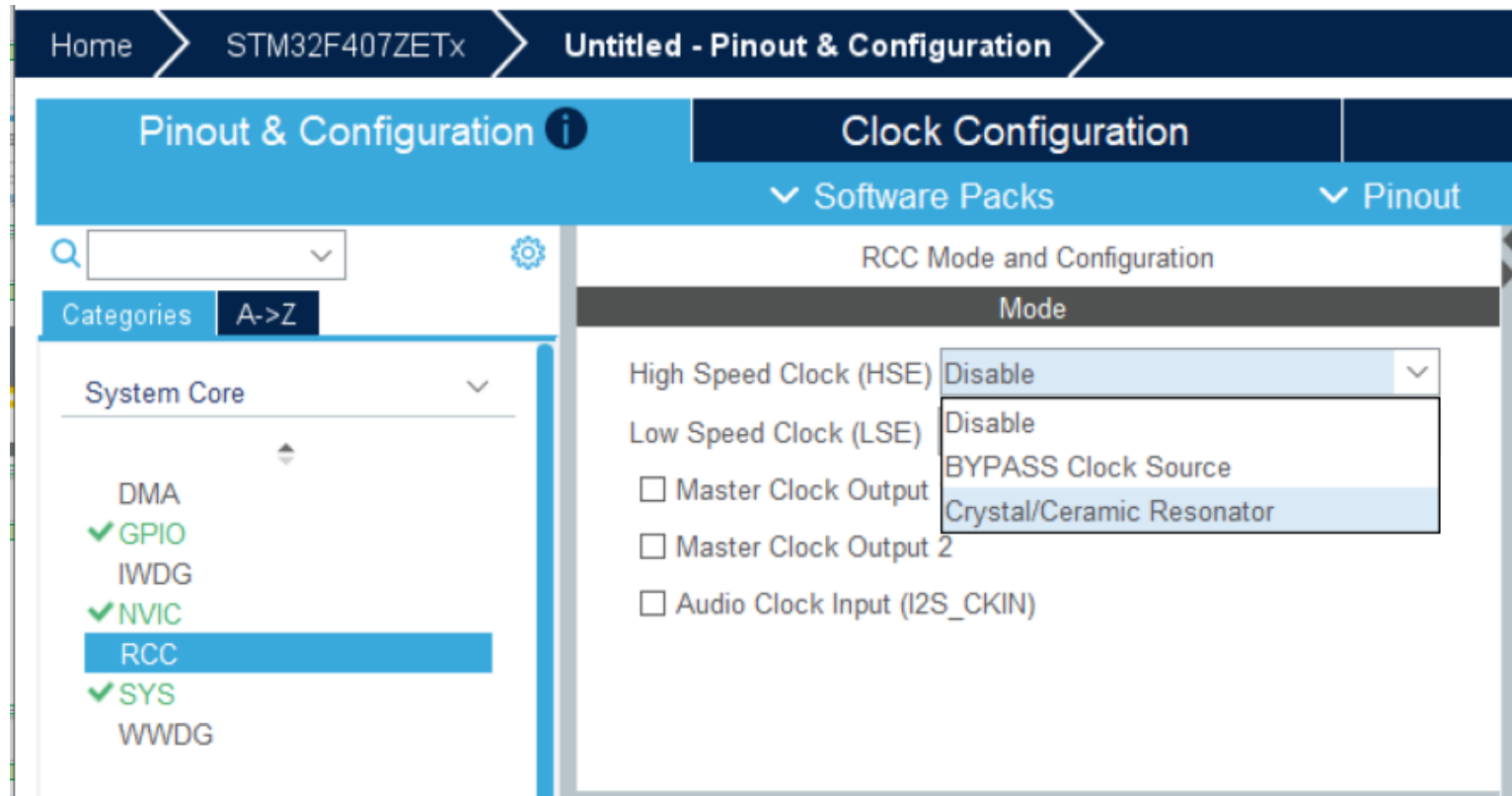
- I2C 핀 설정 - 회로도 참조
- I2C : PB6-I2C1_SCL, PB7-I2C1_SDA

PB5	136	I2C1_SCL
PB6	137	I2C1_SDA
PB7	139	CAN1_RX



Project 생성

- System Core -> RCC에서 HSE 활성화 -> Crystal/Ceramic Resonator 선택



Project 생성

- Connectivity -> USART1 -> Asynchronous 선택

The screenshot shows the STM32CubeMX software interface. The top navigation bar includes 'Home', 'STM32F407ZETx', and 'Untitled - Pinout & Configuration'. The main window is titled 'Pinout & Configuration' and features a left sidebar with a search bar and a list of categories. Under the 'Connectivity' category, various peripherals are listed, with 'USART1' highlighted in blue. The right pane, titled 'Clock Configuration', shows the 'USART1 Mode and Configuration' section. A dropdown menu for 'Mode' is open, displaying options: 'Disable', 'Asynchronous' (selected), 'Synchronous', 'Single Wire (Half-Duplex)', 'Multiprocessor Communication', 'IrDA', 'LIN', and 'SmartCard'. Below this, a 'Configuration' section is partially visible.

Home > STM32F407ZETx > Untitled - Pinout & Configuration

Pinout & Configuration ⓘ

Clock Configuration

▼ Software Packs ▼

USART1 Mode and Configuration

Mode

Mode Disable

Hardware Disable

Asynchronous

Synchronous

Single Wire (Half-Duplex)

Multiprocessor Communication

IrDA

LIN

SmartCard

Configuration

Categories A->Z

Timers >

Connectivity ▼

CAN1

CAN2

ETH

FSMC

I2C1

I2C2

⚠ I2C3

SDIO

SPI1

SPI2

SPI3

UART4

UART5

USART1

USART2

USART3

USART6

⚠ USB_OTG_FS

USB_OTG_HS

Project 생성

- Connectivity -> I2C1 -> Mode -> I2C 선택

The screenshot shows the STM32CubeMX interface for configuring the I2C1 peripheral. On the left, the 'Connectivity' category is expanded, and 'I2C1' is selected. The main window is titled 'I2C1 Mode and Configuration' and is divided into two main sections: 'Mode' and 'Configuration'.

Mode Section:

- A dropdown menu shows 'I2C' selected.

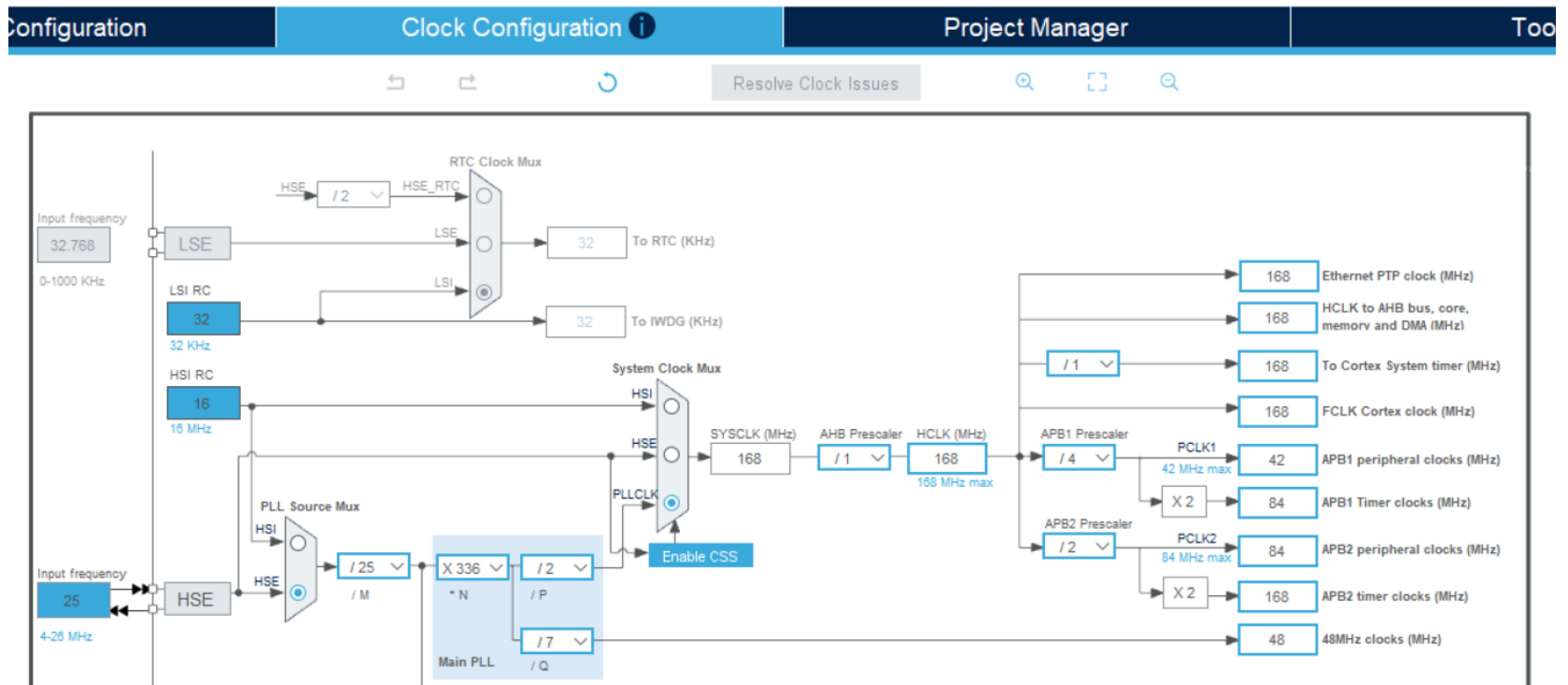
Configuration Section:

- A 'Reset Configuration' button is present.
- Four tabs are visible: 'NVIC Settings', 'DMA Settings', 'GPIO Settings', and 'Parameter Settings' (which is currently selected).
- Below the tabs, a text box says 'Configure the below parameters :'.
- A search bar with the placeholder 'Search (Ctrl+F)' is located above the parameter list.
- The parameters are organized into two sections: 'Master Features' and 'Slave Features'.

Feature	Value
Master Features	
I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000
Slave Features	
Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

Project 생성

- Clock Configuration 클릭 -> Yes 클릭



Project 생성

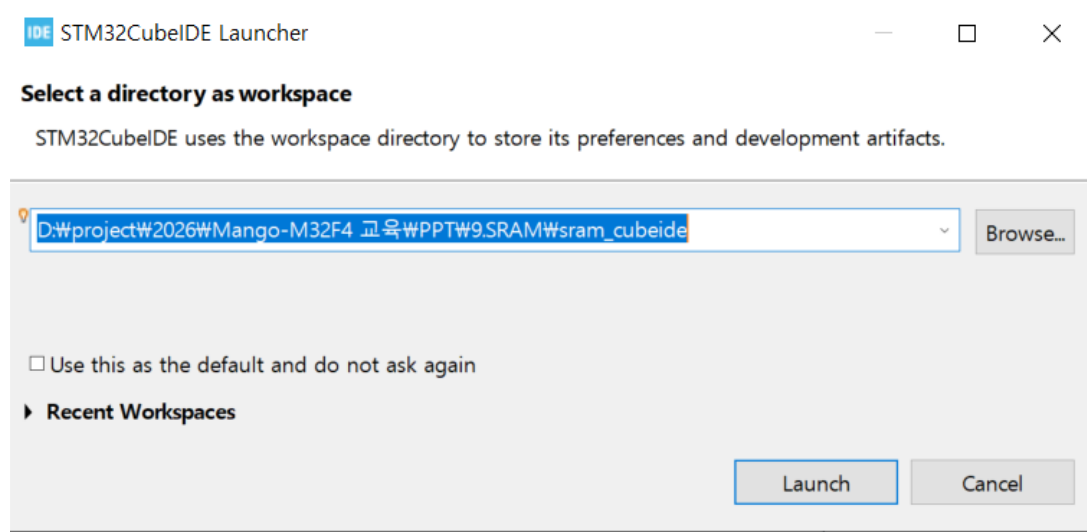
- Project Manager에서 Project Name 기입, IDE를 STM32CubeIDE로 선택 후 GENERATE CODE 클릭

The screenshot shows the STM32CubeIDE Project Manager interface. The top navigation bar includes 'Home', 'STM32F407ZETx', 'Untitled - Project Manager', and a 'GENERATE CODE' button. Below this, there are four tabs: 'Pinout & Configuration', 'Clock Configuration', 'Project Manager' (which is active), and 'Tools'. On the left side of the 'Project Manager' tab, there are two sub-tabs: 'Project' and 'Code Generator'. The 'Project' sub-tab is selected, displaying the 'Project Settings' section. This section contains the following fields and options:

- Project Name:** A text input field containing 'sram_cubemx'.
- Project Location:** A text input field containing 'D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\'. To the right of the field is a 'Browse' button.
- Application Structure:** A dropdown menu set to 'Advanced'. To the right of the dropdown is a checkbox labeled 'Do not generate the main()' which is currently unchecked.
- Toolchain Folder Location:** A text input field containing 'D:\project\2026\Mango-M32F4 교육\PPT\9.SRAM\'. To the right of the field is a 'Browse' button.
- Toolchain / IDE:** A dropdown menu set to 'STM32CubeIDE'. To the right of the dropdown is a checked checkbox labeled 'Generate Under Root'.

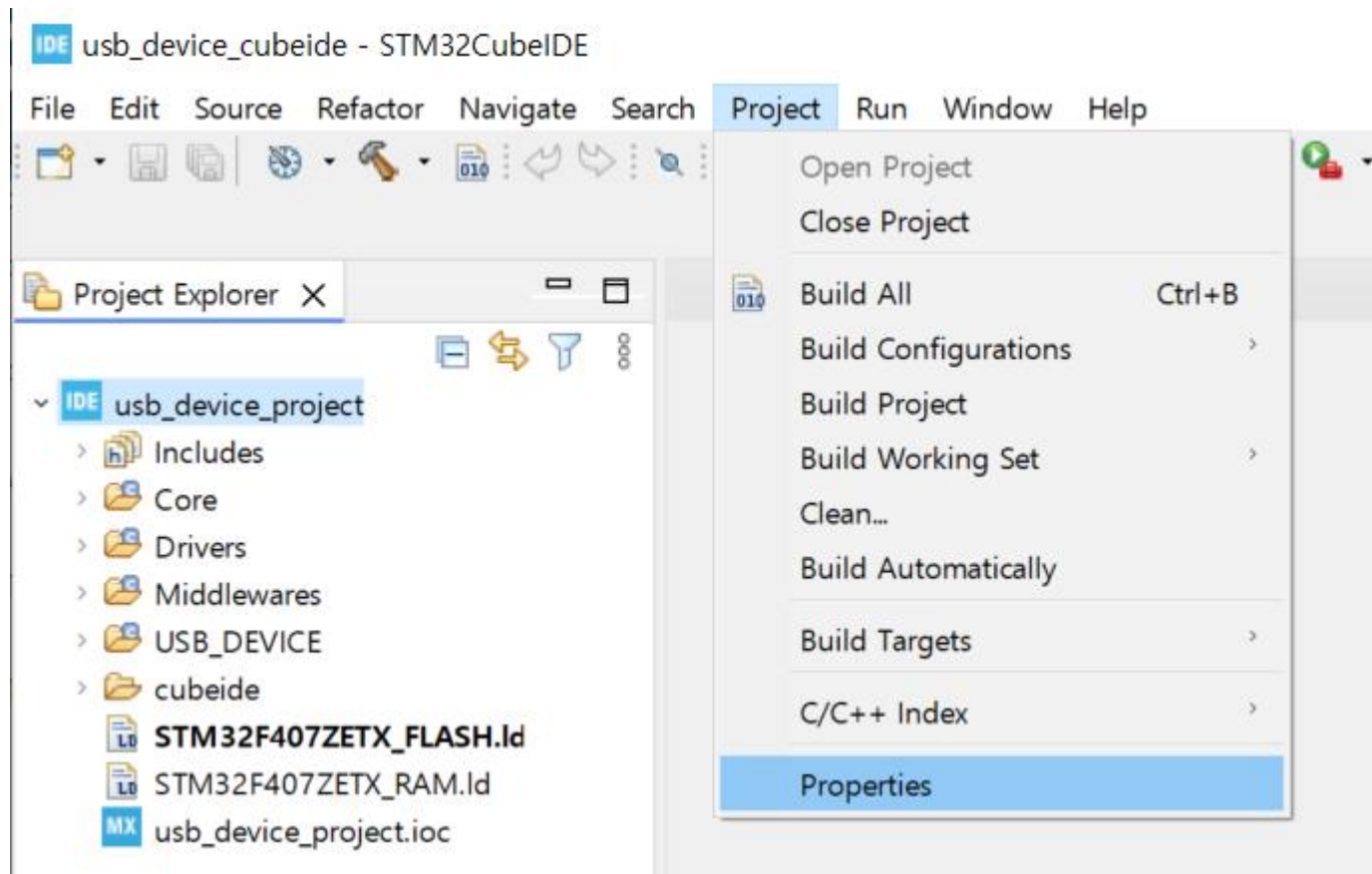
Project 생성

- Open Project 클릭하여 STM32CubeIDE 실행, workspace는 CubeMX와 다른 폴더로 지정



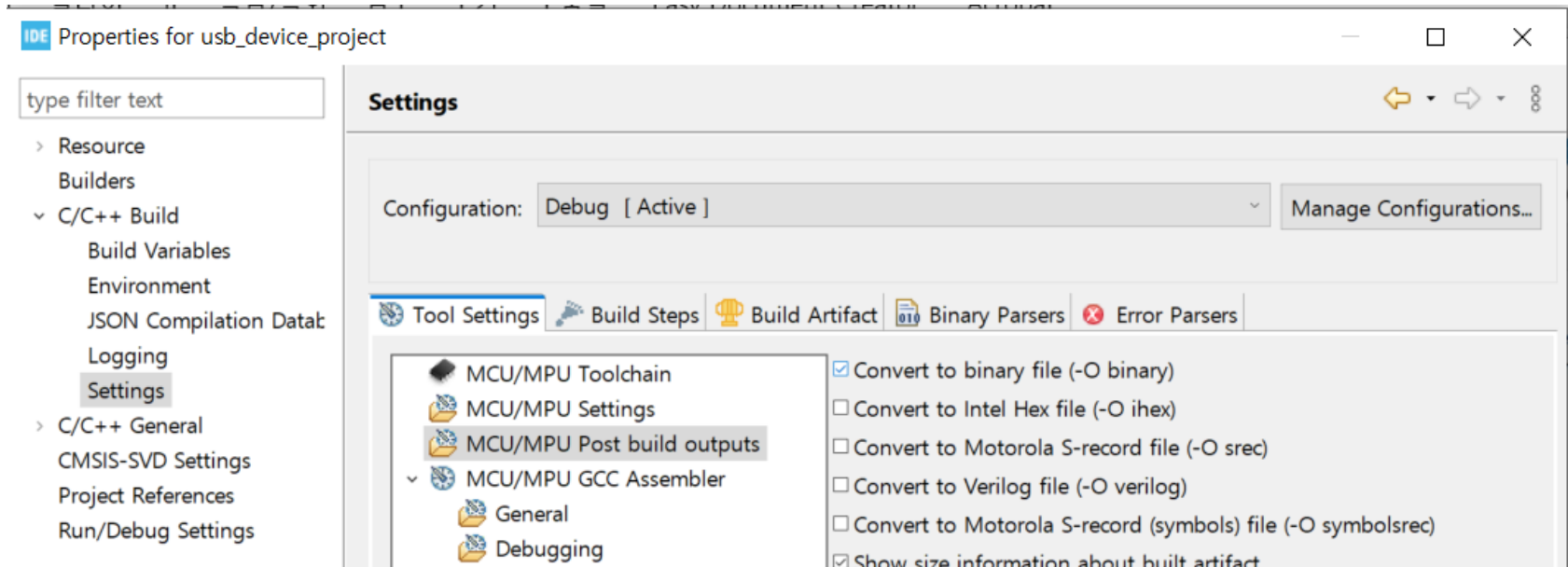
BIN 파일 생성 설정

- STM32CubeProgrammer로 시리얼 다운로드하기 위해서는 BIN파일이 필요.
- Project -> Properties 클릭



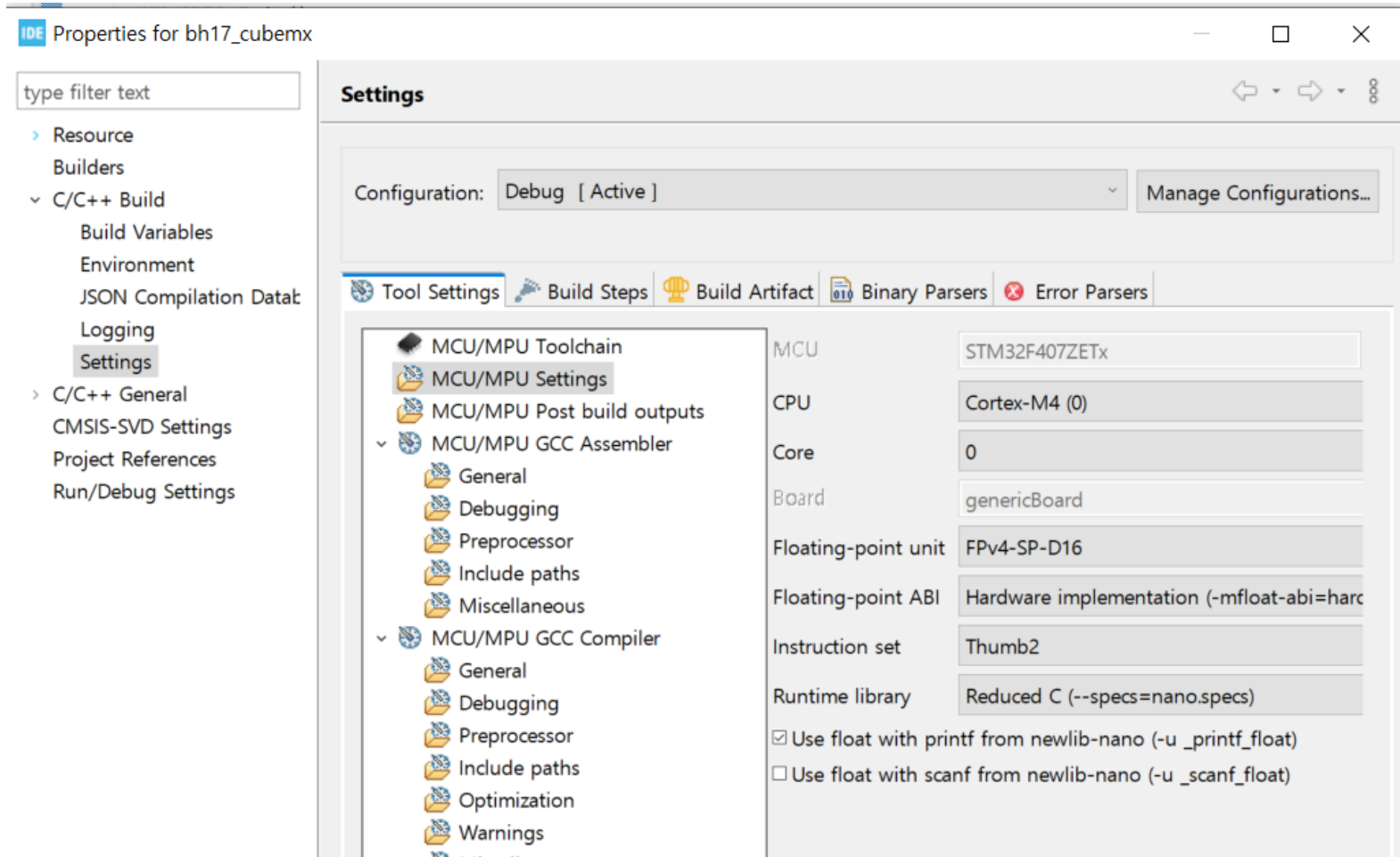
BIN 파일 생성 설정

- C/C++ Build -> Settings -> Tool Settings -> MCU/MPU Post build outputs => Convert to binary file(-O binary) 클릭.



빌드 설정

- C/C++ Build -> Settings -> Tool Settings -> MCU/MPU Settings => Use float with printf from newlib-nano(-u _printf_float) 클릭.



- Slave Address : 0x40

After sending the Start condition, the subsequent I²C header consists of the 7-bit I²C device address '1000'000' and an SDA direction bit (Read R: '1', Write W: '0'). The

- Command Table

Command	Comment	Code
Trigger T measurement	hold master	1110'0011
Trigger RH measurement	hold master	1110'0101
Trigger T measurement	no hold master	1111'0011
Trigger RH measurement	no hold master	1111'0101
Write user register		1110'0110
Read user register		1110'0111
Soft reset		1111'1110

Table 6 Basic command set, RH stands for relative humidity, and T stands for temperature

- Humidity Conversion

- **Relative Humidity conversion**

With the relative humidity signal output S_{RH} , the relative humidity is obtained by the following formula (result in %RH), no matter which resolution is chosen:

$$RH = -6 + 125 \times \frac{S_{RH}}{2^{16}}$$

- Temperature Conversion

- **Temperature conversion**

The temperature T is calculated by inserting temperature signal output S_{Temp} into the following formula (result in °C), no matter which resolution is chosen:

$$Temp = -46.85 + 175.72 \times \frac{S_{Temp}}{2^{16}}$$

Code 수정

- UART1으로 로그를 내 보내기 위해 아래 코드 추가 :
core/src/main.c

```
/* Private user code -----*/
/* USER CODE BEGIN 0 */
#define hUART huart1
int _write( int32_t file , uint8_t *ptr , int32_t len )
{
    /* Implement your write code here, this is used by puts and printf for example
    */
    for ( int16_t i = 0 ; i < len ; ++i )
    {
        HAL_UART_Transmit( &hUART, ptr++, 1, 100);
    }
    return len;
}
/* USER CODE END 0 */

/**
 * @brief The application entry point.
 * @retval int
 */
int main(void)
{
```

- SHT20의 slave address는 0x40이다.

```
/* USER CODE BEGIN PD */
```

```
#define SHT20_I2C_ADDR (0x40 << 1)
```

```
#define REG_RTEMP_READ_HOLD      0xE3
```

```
#define REG_RHUM_READ_HOLD      0xE5
```

```
#define REG_SOFT_RESET          0xFE
```

```
//CRC error macros
```

```
#define CRC_MATCH                1
```

```
#define CRC_NOTMATCH             0
```

```
//CRC macros
```

```
#define CRC_POLYNOMIAL           0x131 // Generator Polynomial  $X^8 + X^5 + X^4 + 1$ 
```

```
#define CRC_POLYNOMIAL_WIDTH     9
```

```
/* USER CODE END PD */
```

- CRC 함수.

```
static int crc_check(uint16_t dt, uint8_t crc)
{
    uint16_t polynomial = CRC_POLYNOMIAL << (16 - CRC_POLYNOMIAL_WIDTH);
    uint16_t result = dt;
    uint16_t top_bit = 0x8000;

    for(int i=0;i<16;i++)
    {
        if(result & top_bit) result ^= polynomial;
        result <<= 1;
    }

    result >>= 8;

    return crc == result ? CRC_MATCH : CRC_NOTMATCH;
}
/* USER CODE END 0 */
```

- 초기화시에 Software Reset를 한다.

```
int main(void)
{
    uint8_t reg;
    uint8_t data[3];
    uint16_t data_raw;
    uint8_t crc;
    int raw_value;
    HAL_StatusTypeDef status;

    printf("I2C-SHT20 test\r\n");

    HAL_Delay(1000);
    reg = REG_SOFT_RESET;
    status = HAL_I2C_Master_Transmit(&hi2c1, SHT20_I2C_ADDR, &reg, 1, 1000);
    HAL_Delay(300);
```

Code 수정

- 1초 주기로 온도와 습도 정보를 표시한다.

```
while (1)
{
    HAL_Delay(1000);
    reg = REG_RTEMP_READ_HOLD;
    status = HAL_I2C_Master_Transmit(&hi2c1, SHT20_I2C_ADDR, &reg, 1, 1000);
    status = HAL_I2C_Master_Receive(&hi2c1, SHT20_I2C_ADDR, data, 3, 1000);
    data_raw = (data[0] << 8) | data[1];
    crc = data[2];
    if(crc_check(data_raw, crc) == CRC_MATCH)
    {
        data_raw &= 0xFFFC;
        raw_value = ((17572 * data_raw) >> 16) - 4685;

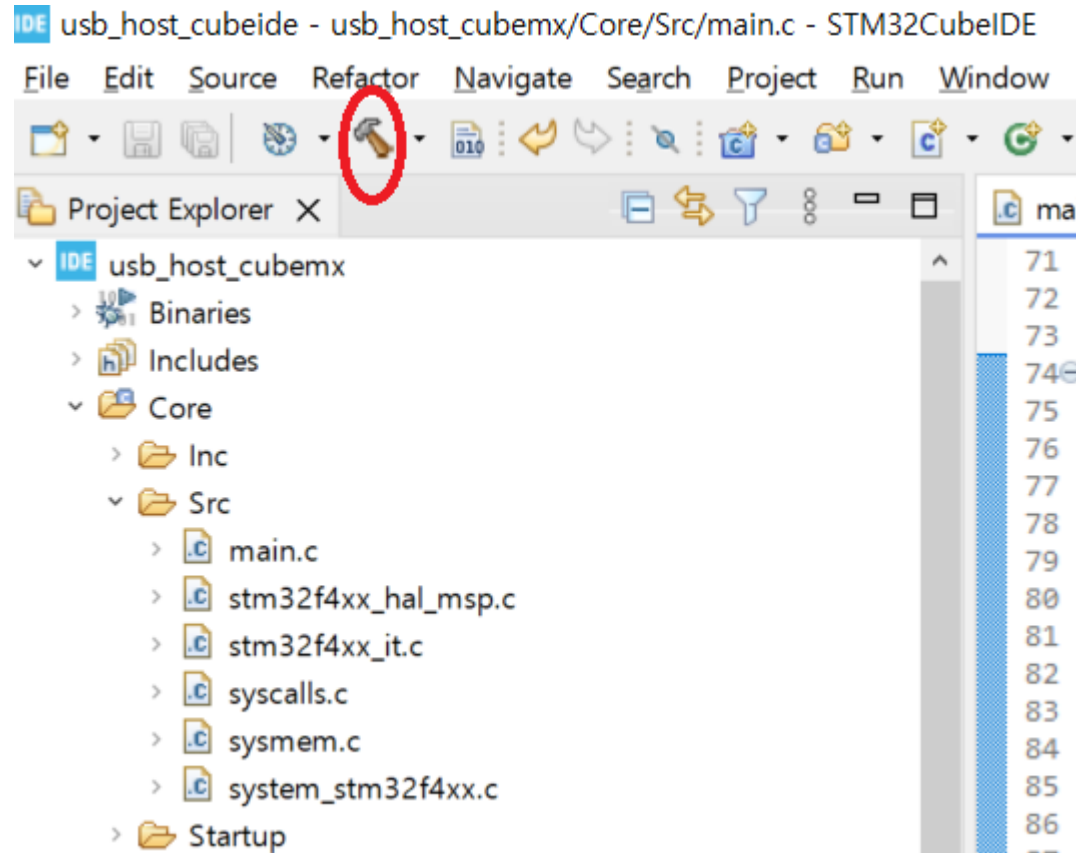
        printf("TEMP=%d.%d°C\r\n", raw_value / 100, raw_value % 100);
    }

    reg = REG_RHUM_READ_HOLD;
    status = HAL_I2C_Master_Transmit(&hi2c1, SHT20_I2C_ADDR, &reg, 1, 1000);
    status = HAL_I2C_Master_Receive(&hi2c1, SHT20_I2C_ADDR, data, 3, 1000);
    data_raw = (data[0] << 8) | data[1];
    crc = data[2];
    if(crc_check(data_raw, crc) == CRC_MATCH)
    {
        data_raw &= 0xFFFC;
        raw_value = ((125 * data_raw) >> 16) - 6;

        printf("HUMID=%d%%\r\n", raw_value);
    }
}
```

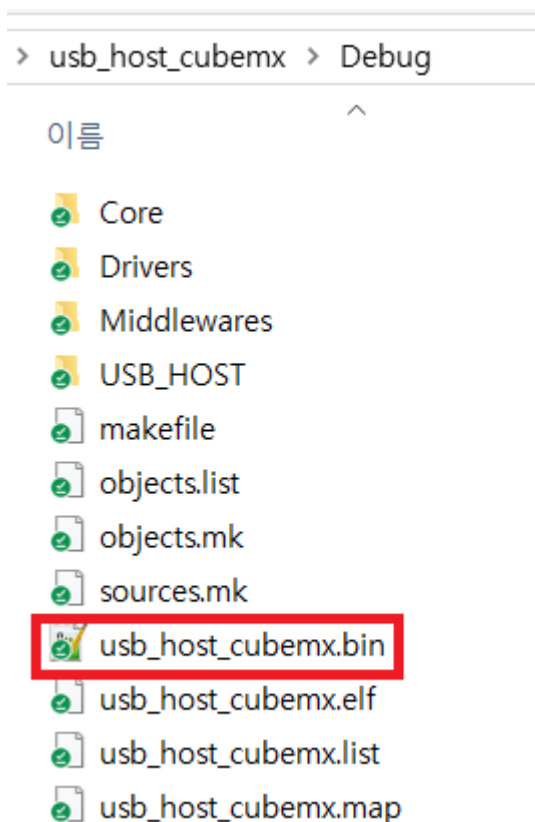
Build

- 아래 버튼 클릭하여 Build



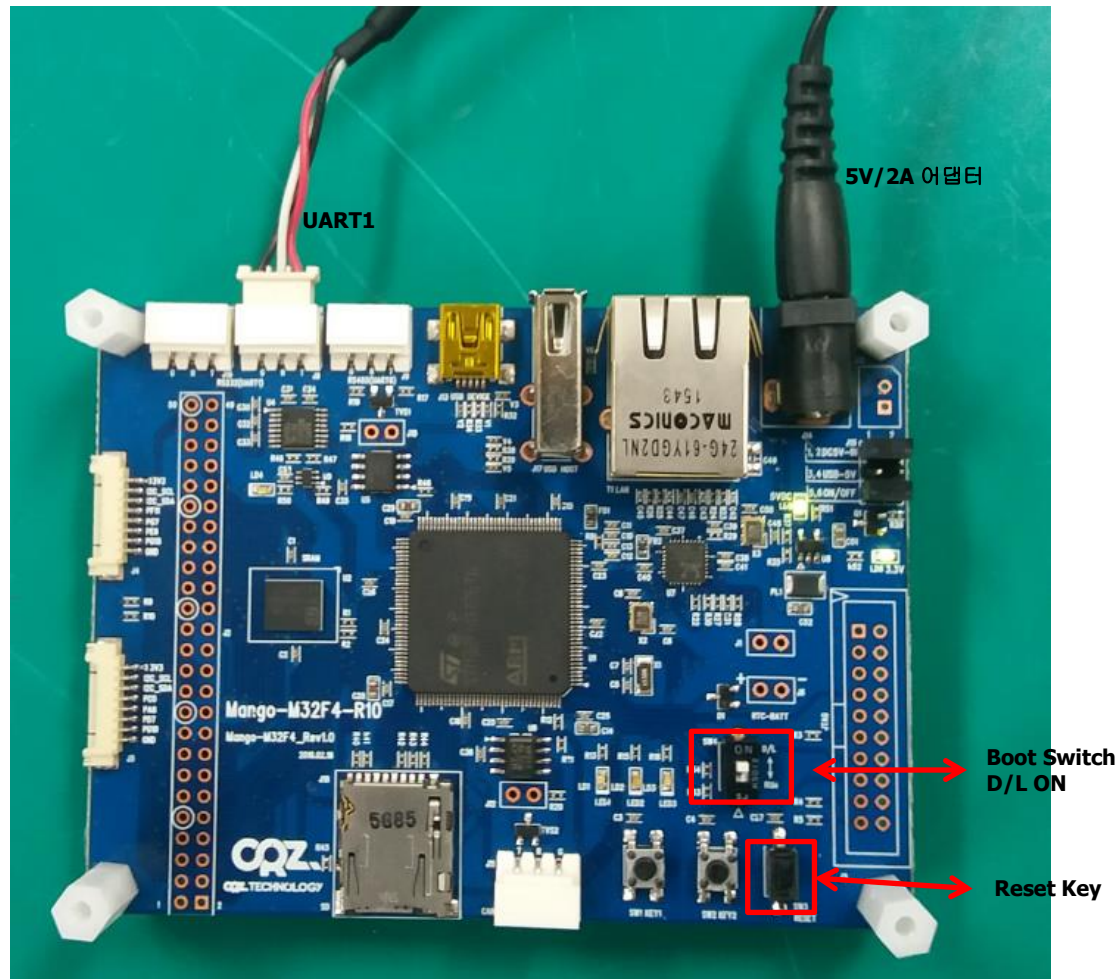
Build 결과물

- 빌드 결과물 이름은 [프로젝트 이름].bin이다.



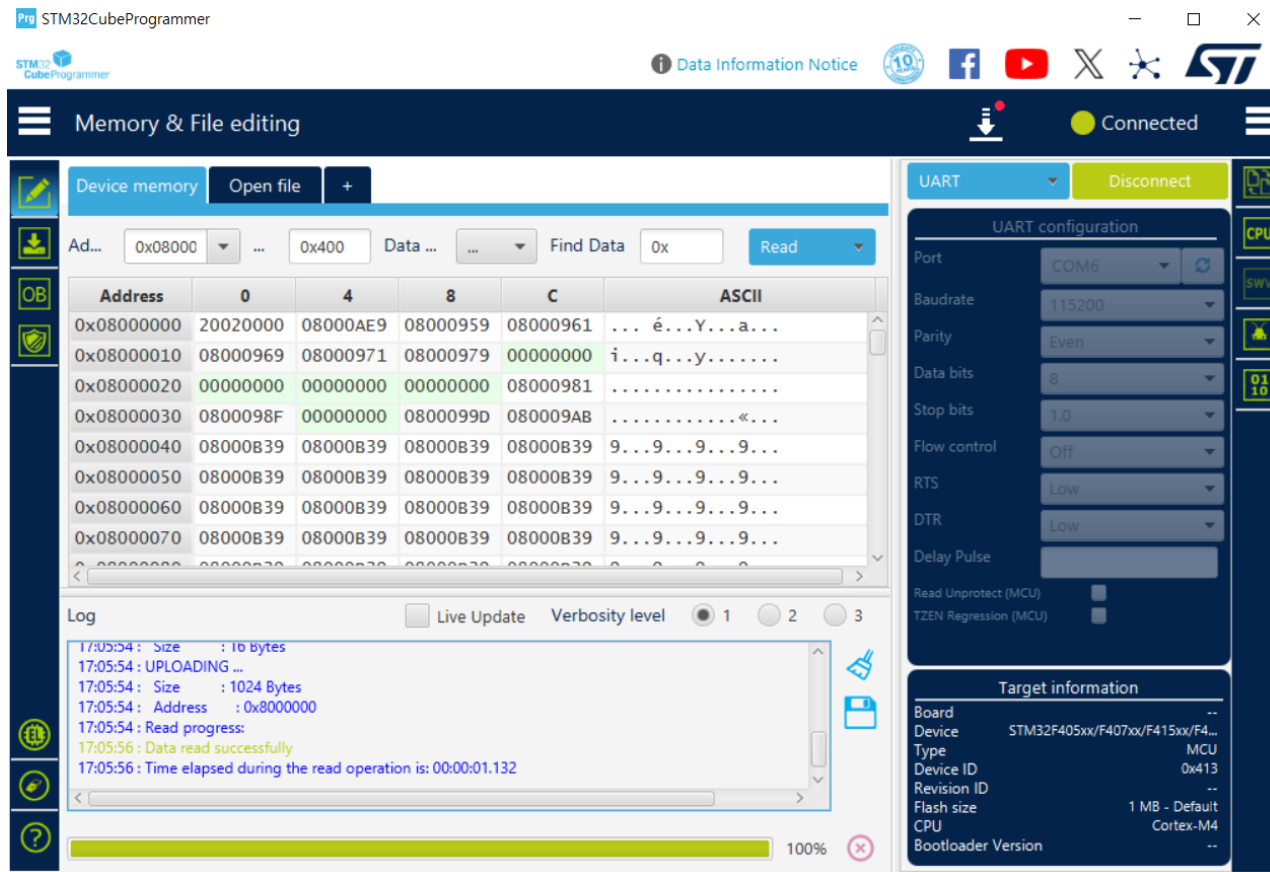
BIN 다운로드

- UART1과 연결되어 있는 터미널(Teraterm) 종료.
- Mango-M32F4를 다운로드 모드로 설정하고 SW3 리셋 키를 누른다.



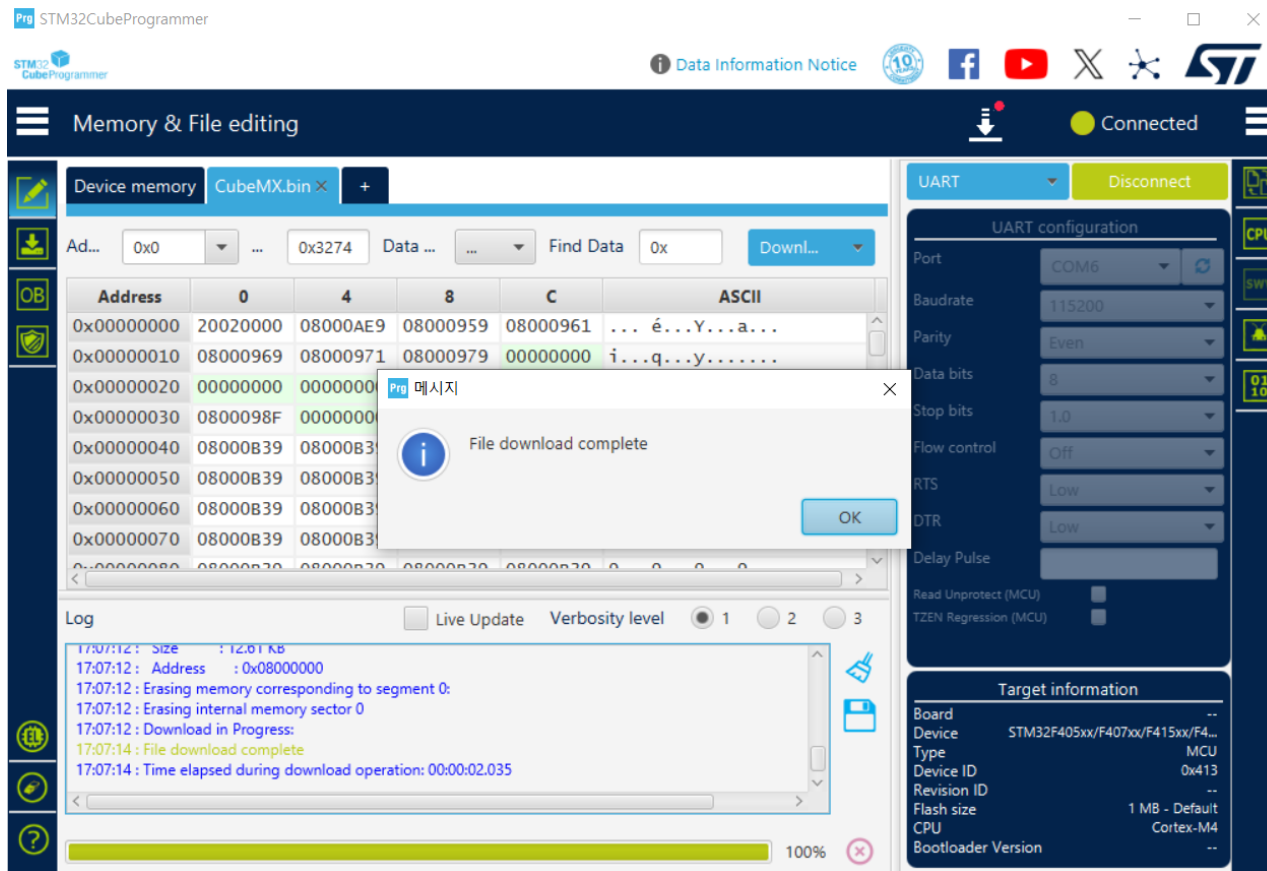
BIN 다운로드

- STM32CubeProgrammer를 실행한다.



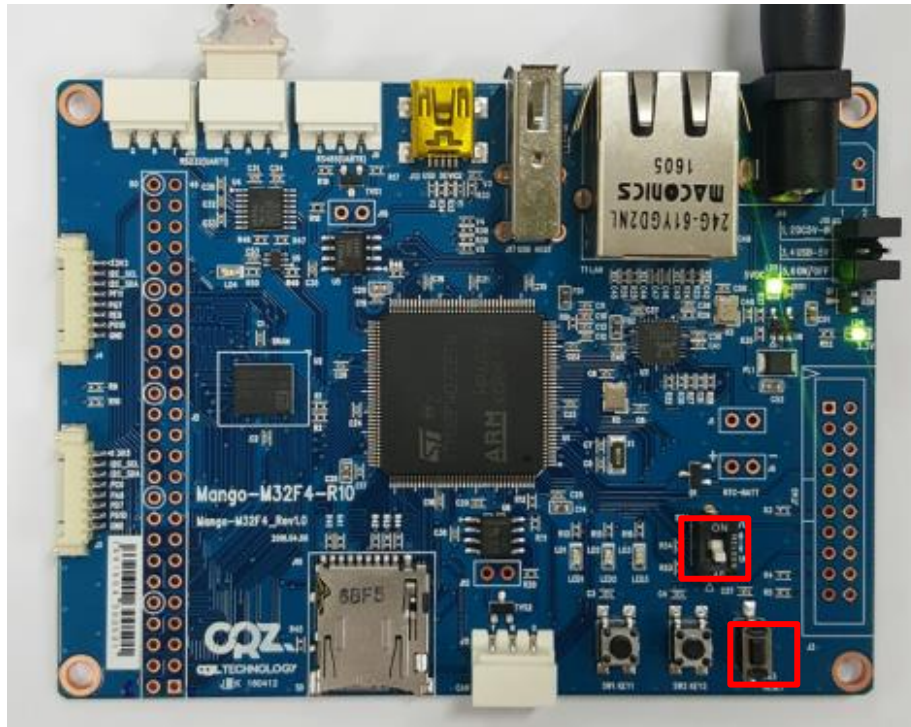
BIN 다운로드

- 빌드된 BIN 파일을 열어서 다운로드한다.



실행결과

- STM32CubeProgrammer의 COM Port를 연결해제하고 터미널 (Teraterm)을 실행한다.
- Mango-M32F4를 실행모드로 설정하고 SW3 리셋 키를 누른다.



실행결과

- CX-HUMIDITY 센서 보드를 Mango-M32F4 보드의 J4 혹은 J5에 8핀 커넥터를 이용하여 연결한다.
- 온도와 습도가 정상적으로 읽힘.

A screenshot of a Tera Term VT window titled 'COM6 - Tera Term VT'. The window displays a series of sensor readings from an I2C-SHT20 sensor. The readings are as follows:

```
I2C-SHT20 test
TEMP=25.14 °C
HUMID=41%
TEMP=25.15 °C
HUMID=41%
TEMP=25.15 °C
HUMID=41%
TEMP=25.15 °C
HUMID=41%
TEMP=25.15 °C
HUMID=41%
TEMP=25.15 °C
HUMID=41%
TEMP=25.15 °C
HUMID=41%
TEMP=25.14 °C
HUMID=41%
TEMP=25.14 °C
HUMID=41%
TEMP=25.14 °C
HUMID=41%
TEMP=25.13 °C
HUMID=41%
```

The window has a menu bar with 'File', 'Edit', 'Setup', 'Control', 'Window', and 'Help'. The background is black with white text.